SOLO BATTERY MOD FOR DUMMIES

(Brought to you by JB's (Just Better) 3DR Solo Owners Group)

PART ONE: PREPARING THE BATTERY FOR BMS CALIBRATION

WARNING!





Now that I have your attention PLEASE take heed to the following DISCLAIMER:

LiPOs, especially those that have reached the end of their lifespan are VERY unstable and even the slightest nick through the outside jacket can be CATASTROPHIC. There is a very real element of HIGH RISK of injury when performing the following battery mod. Accept this risk or walk away. If you are in need of a fresh battery and not comfortable with the risk involved or after you read through this procedure you feel you don't posses the skills-base to DIY look to the end of this tutorial for several alternatives to powering the Solo.

There are several excellent "Battery Mods" as they are referred to, some are excellent and some not so much. After spending time reviewing most of them I swept them all up in a pile and gleaned the best points to come up with this procedure.

STEP ONE- BMS EXTRACTION:

So what exactly IS a “BMS”?

A **battery management system** (**BMS**) is any electronic system that manages a rechargeable battery (cell or battery pack), such as by protecting the battery from operating outside its safe operating area, monitoring its state, calculating secondary data, **reporting that data**, controlling its environment, authenticating it and / or balancing it.

The jury is still out on a verdict but conventional wisdom leans on the side of every BMS is married to one and only one battery. There are a lot discussion wrapped around that statement. “Hot swapping” the BMS comes at a risk of not having the correct information streamed to the autopilot. **This incorrect data can result in delayed fail-safe warnings.** A delayed RTL can result in the Solo not making it back home on it’s remaining battery capacity. This is our presentation and I’m going with the one BMS to One Battery mindset.

Solo as do most Prosumer and Commercial UAVs utilize “Smart” Battery Packs. Besides the high cost what makes a smart battery smart is the BMS. A very important duty of the Solo BMS is to stream data back to the Controller which displays useful information on the apps HUD. Additionally this data is sent to the auto pilot and compared to a burned in set of flight parameters. These parameters are what warn you to a low battery and at a preset point when the remaining capacity of the battery falls at which point a fail-safe triggers RTL (Return to Land). When things really get desperate and you are smart enough to be flying the Solex app you will hear the dreaded “Credit Card Lady” Announcement.

That’s why we need to extract the BMS from the bricked battery. Without it on board you get no telemetry. No telemetry no HUD data and MOST important no Battery Fail-safe RTL.

WEAR FACE PROTECTION! GOGGLES ARE A MUST - BEST IS A FULL FACE MASK

To extract the BMS we need to separate the top and bottom battery case. If you forgot the message I started out with, go back now and watch that video.

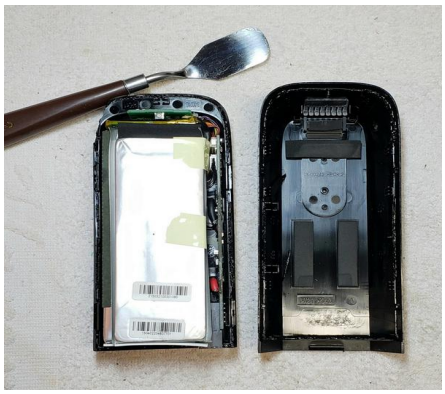
THERE IS NO SAFE METHODOLOGY THAT CAN BE APPLIED TO SPLITTING THE TWO BATTERY CASE HALVES which is why you are now wearing face protection. That said the ONLY (safest) method to be applied to the task is by applying a very light (drops) application of Glue Buster to the seam between the two halves. BUY GLUE BUSTER







After running a tiny bead of the Glue Buster all around the seam between the two cases let it sit for about 5 minutes. Then (very carefully if you plan on salvaging the cases for reuse) using a butter knife, screwdriver, artists scapula, putty knife, or any combination of the above, pry the cases apart going around the seam several times. Also break the bond between the bottom of the battery and the case as shown in the photo above. Glue Buster won't hurt you but it does mix with the glue which devolves it and all but destroys the plastic case. The top case is made of a different composition than the bottom case and if you are very gentle in your prying method you can salvage at least the top section as it seems not not be as effected by the aforementioned chemical reaction.
UPDATE! After applying the Glue Buster you only wait a few seconds before separating the cases and then as soon as you do apply a damp cloth to the area which will stop the reaction and save both sections.



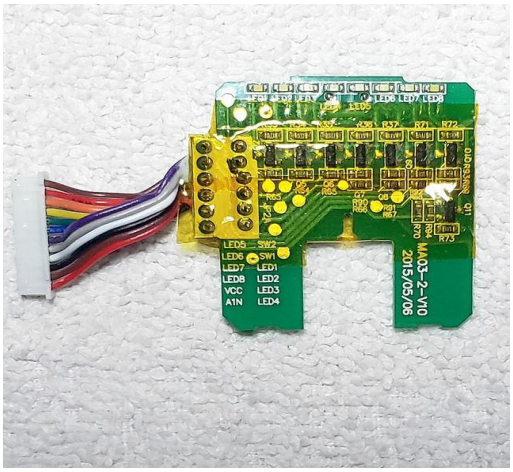


Once you have separated the cases set them out of the way and lifte the innards out.
CONGRATULATIONS! If you are still with me you haven't blown yourself up….yet.

STEP TWO- REMOVE BMS FROM SOLO BATTERY

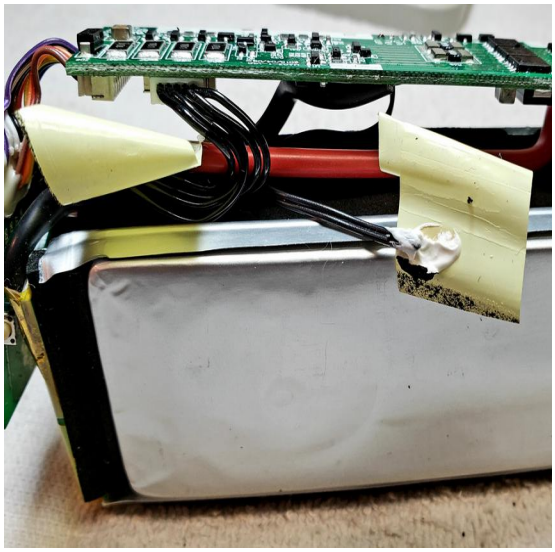
DO NOT REMOVE YOUR FACE PROTECTION!

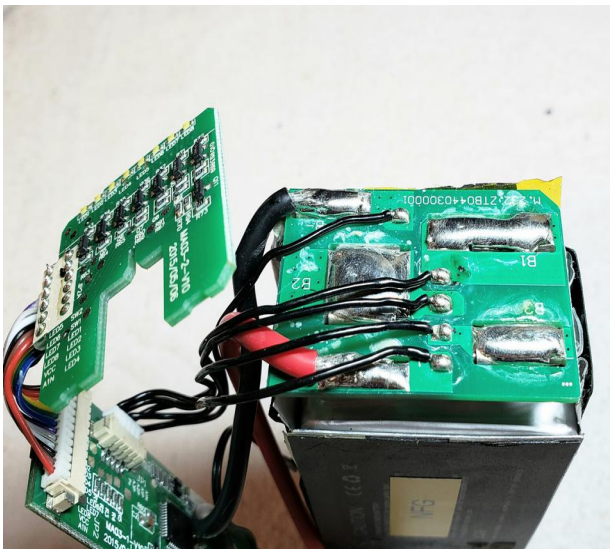
First and foremost before you go any further you must protect the exposed terminal taps on the both the LED and the BMS printed circuit boards. Best to use ESD tape as it is resistant to heat, static, and is an excellent insulator. One more precautionary measure is to dab some epoxy glue behind the power button. It wasn't designed to be fat fingered and will easily dislodge from the board. Do NOT use super-glue as it has been documented to seep into the switch and glue the button fast.



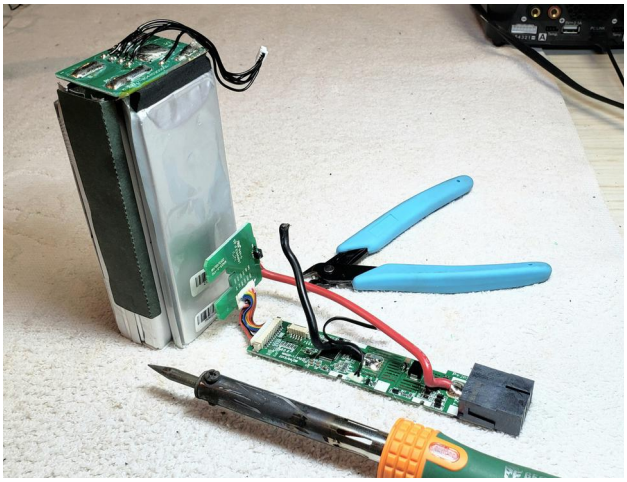


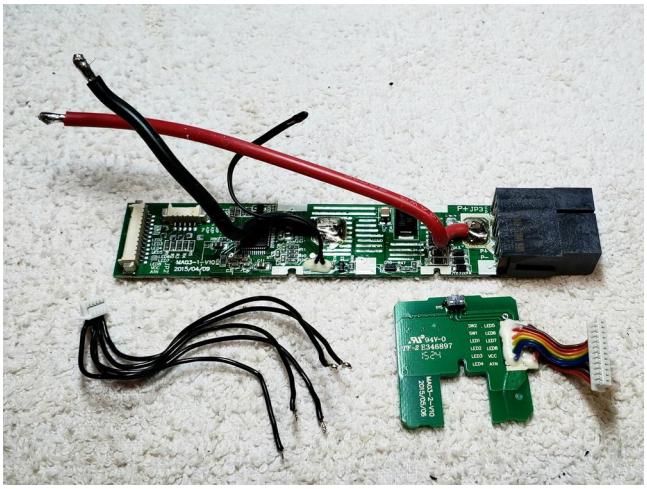
Next set the battery on it's side and gently peel, not cut, the tape holding the power leads and thermistor (thing with the glob of silicone on it that reads the temp of the battery) I used a wooden stick. Remember you cut through the very thin cell wrapper you go **POOF**! Next carefully expose the terminal board on the battery taking EXTREME care not to arc across any of the now exposed wire connections.





At this point I'll give you a choice of snipping off the five balance wires and power leads or unsoldering them. I prefer to unsolder them as it provides well tinned wire ends to attach to the new battery pack. Again, take caution not to arc the terminals. Cut the wires one at a time.





PROPERLY DISPOSE OF THE OLD BATTERY!

Option 1 - Here's a tried and true method-

1. Get a plastic container which you can afford to throw away, fill it with cold water
2. Mix it with salt and make sure it's dissolved completely. Use about 2 Tablespoons of salt per quart of water.
3. Put the battery in the salty water, and leave it somewhere fire proof for two weeks. Depending how much charge it had originally, you might want to leave it longer.
4. Finally, wrap the battery with paper and it's ready to be thrown away.

Option 2 - Recycle it: tape over the connector and any exposed terminals, keep it in a fireproof container, and take it to a battery recycling drop-off. Many hardware and electronics stores accept lithium batteries.

A QUICK WORD ABOUT REPLACEMENT BATTERIES-

The OEM battery pack (what you now have sitting on your dining room table less the BMS) was manufactured by Gens Ace Tattu. Original specs were a 16.80V (Peak voltage output) 5200mAh (Battery Storage Capacity) 10C (Current output) four cell pack. There are several manufactures of LiPO batteries out there. As you would expect they come in all shapes and sizes. Up until the middle of November 2018 Tattu offered a battery that met all of requirements in size, weight, capacity and power output. Actually the battery I use in this presentation exceeds the current output of the OEM battery (10C OEM vs 35C Tattu Replacement). This allows the battery to run much cooler and thereby extend battery life.

Take note that a higher Current Output does not improve flying performance with the exception of flat out Acro mode which few of us fly. The OEM 10C Pack was adequate for all but the most severe Acro maneuvers but it still got hot. Actually there have been cases where overload in Acro resulted in the Pixhawk shutting down power. Hence UAV becomes a rock. Having a higher current draw capability is what results in better efficiency and cooler running.

Currently there are several battery packs that will make a suitable alternative to perform the Battery Mod. For the purposes of this tutorial and based on our experience with the GensAce Tattu in general I highly recommend you go with the Tattu unit.

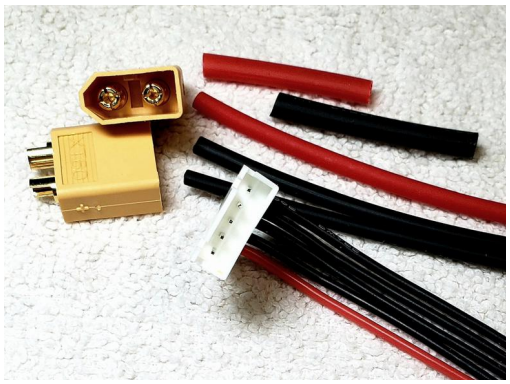
If you don't have access to these packs and need to consider an alternative then be mindful of these dimensions and weight in addition to the packs output specs. These are the specs for the Tattu batteries that we know will work-
4S1P/14.8V/5200mAh/35-40C Size: 133m X 45m X 33m Weight: 436g

STEP THREE- PREPARE REPLACEMENT BATTERY FOR BMS ATTACHMENT

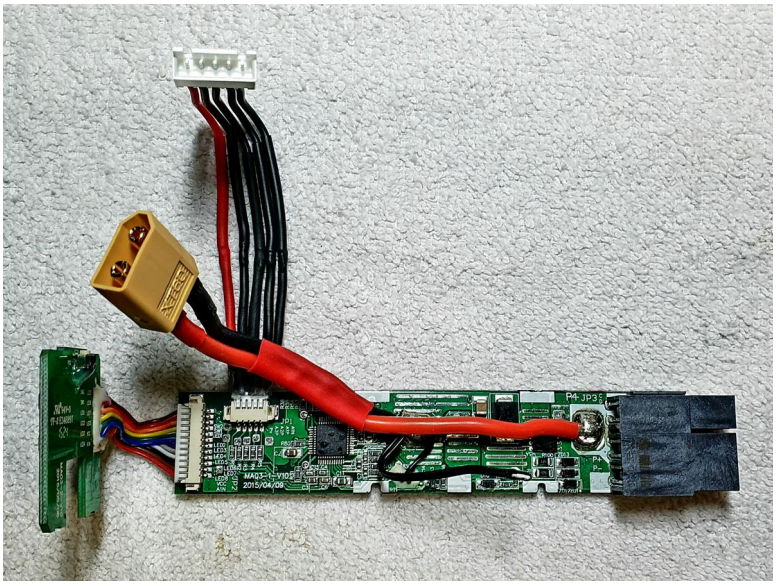
DO NOT REMOVE YOUR FACE PROTECTION!

There are two popular methods to connect the BMS you extracted in step two. One is to attach connectors to the BMS so that it is readily connected to the power and balance connectors on the replacement battery. You will need to procure a supply of XT-60 Male Connectors, JST-XH 5-4S Five wire pigtails, and Heat Shrink Tube in assorted diameters.

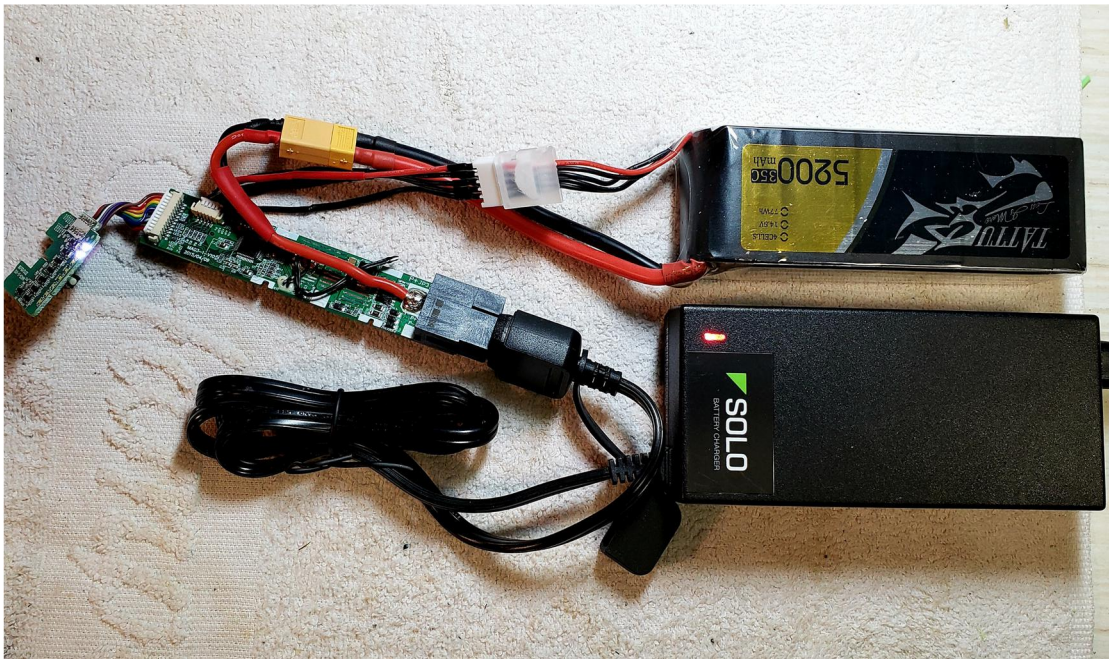
1. NOTE the orientation of the connector blocks in the photo below. Once prepared solder the wires, slide the tubes over your splices, and shrink. A hairdryer on high should do it.
2. Attach the XT-60 connector to the power leads on the BMS. NOTE the red (+) wire attaches to the flat side marked with a + on the connector.







I’m jumping a little ahead showing the charger but this is how the replacement battery looks like connected to the BMS.



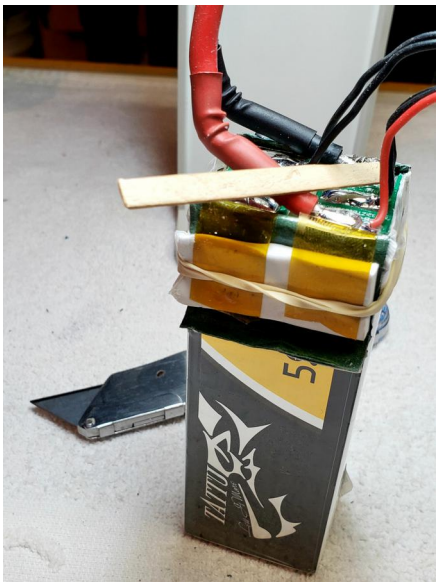
The other popular method to attach the BMS to your replacement battery is to solder the wires directly between the replacement battery and the BMS leads we exposed above.

This method eliminates the need for the connectors and allows more freedom to orient the modded battery in the battery case. This assumes that you will want to reuse the case and that you didn't damage it in the BMS extraction. As we near the end on this presentation I will show you a 3D Printed case that works well.

DO NOT REMOVE YOUR FACE PROTECTION!

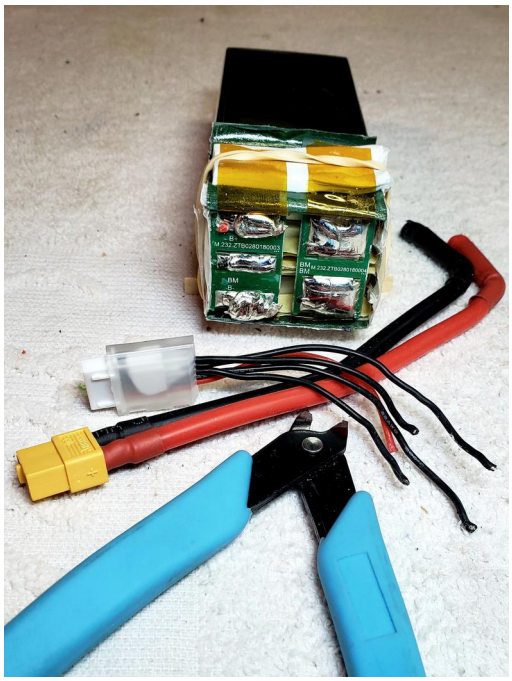
Lets start with the replacement battery. As always you must exercise extreme care in working with a LiPO battery. Cutting through the cell membrane will cause a fire if not explosion. Remember that video.

1. Very carefully cut around the top of the battery to expose the wire taps. Peel back the Polyimide tape and orient the battery as shown in the photo. I use a small stick so as not to short any of the taps.





2. Remove the power leads and balance wires from the battery. You can unsolder them or cut them which ever you are more comfortable with. **CAUTION!** Be very careful not to short out the taps with your soldering gun or side cutters.

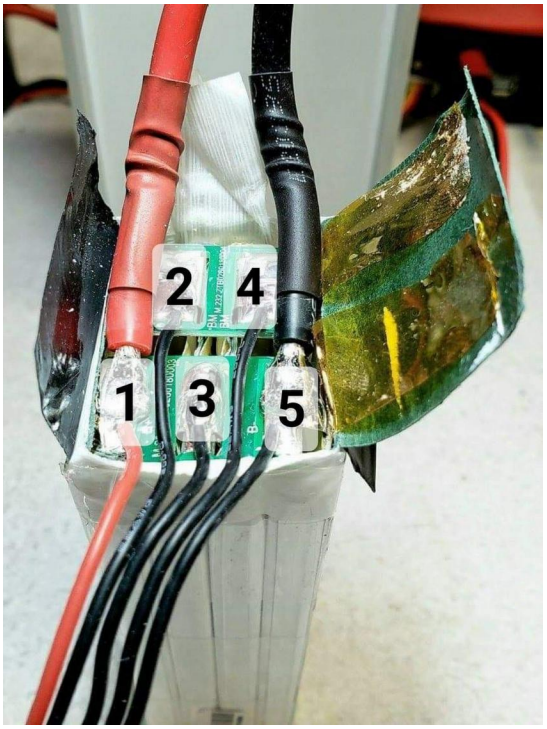


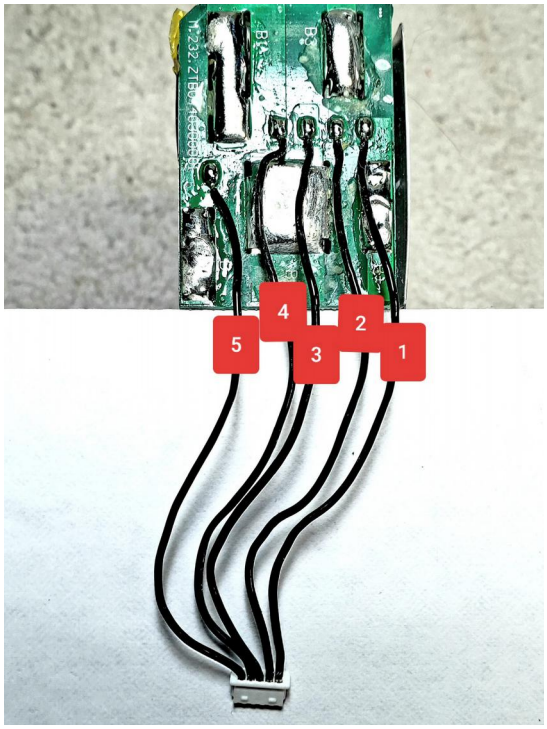
After cutting the wires off the taps I heated and removed the remaining power wire ends off the battery taps to prepare for the BMS power leads to be soldered in their place.

3. Attach the BMS Balance Harness and Power Leads to the replacement battery. Note the number sequence on the replacement battery on the left as it compares to the BMS Harness (still attached to the solo battery) on the right. The #1 wire on the harness terminal from the BMS connects to the #1 tap along with the + power lead on the replacement battery. DON'T MIX THESE UP! They must attach to the replacement order in this sequence.

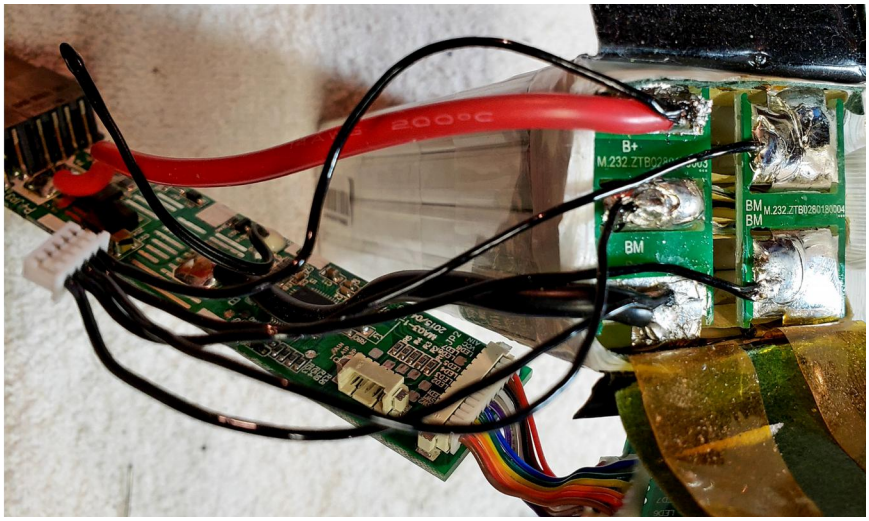
Attach the wires in this order-

1. #1 Balance wire along with the + (Red) Power Lead from the BMS.
2. Balance wires #2, #3, #4
3. After all connections have been made in 1 and 2 then connect Balance wire #5 along with the – (Black) Power lead.



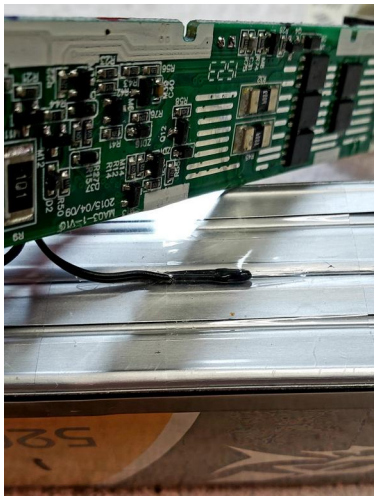


When you are all done it should look like this.-
Do not connect the balance harness to the BMS that will come in good time.



4. Replace the Polyimide tape over the terminals as you found them. One strip over the power terminals and under #2 and #4 balance wires and the other strip over them. Replace the foam strip and cover. Use electrical tape to seal the cover where you cut it loose. CAREFULLY put a small slit in the battery wrap and slide the thermistor in against the battery. Add a piece of foam between the temp probe and board to insulate it from any heat generated by the BMS that can inadvertently cut off the charge cycle. Once all is properly situated you can plug balance harness to the BMS.

DRUM ROLL! Push the power button as you have always done in the past and (hopefully) you will see the normal LED boot up sequence followed by how ever many lights correspond with the charge that was in the replacement battery when you unboxed it.





I wrapped the BMS to the replacement battery with electrical tape followed by a 103 mm wide heat shrink tube.





If you made it this far, you haven't burned down your house, and still have all your fingers and toes then you need to pat yourself on the back. You done good! Now on to the BMS calibration.

Since first putting this out to the Group it was brought to my attention that some considering performing the battery mod might not be up to speed on the soldering aspect of the process. Please be advised that if you have no or little soldering experience I strongly recommend that you DO NOT learn the "How To Solder" with this project. Applying too much heat to the battery terminals can produce disastrous results so consider yourself forewarned. LEARN TO SOLDER VIDEO

PART TWO: CALIBRATING THE BATTERY TO THE BMS

As was the case with preparing the alternative battery for calibration to the BMS you have several choices with how you perform the actual calibration. I'll go over the most popular methods. A few words in general about the WHY and then into the HOW.

Way back in Part One I provided a basic definition of what a BMS does. I'll paste it here so you don't have to back up to read it-

A **battery management system** (**BMS**) is any electronic system that manages a rechargeable battery (cell or battery pack), such as by protecting the battery from operating outside its safe operating area, monitoring its state, calculating secondary data, **reporting that data**, controlling its environment, authenticating it and / or balancing it.

So here in lies the $64,000 question- **How is this accomplished?** (any of you youngsters reading this ask your dad or granddad about the $64,000 question thing)

I won't get into the details just let it suffice that the Solo BMS-

1. "Reads" the individual cell voltage outputs. That's why there are 5 wires in the balance harness you moved over to the replacement battery. One to each cell and common ground.
2. Adds the four cells up and reports the Total voltage.
3. Monitors and reports the total capacity and remaining capacity in mAh (milli Amp hours)
4. Measures and reports the current temperature.
5. Records how many times the battery was cycled
6. Records and reports the current status and condition of the battery.
7. Auto discharges the Battery to 50-54% charge for optimum long term storage.

There are lots of other useful information such as who manufactured the battery and when it was produced to name a few which is burned into the chip's flash memory.

When you connect the replacement battery to the extracted BMS it will usually display the data it recorded from the battery you cut it off of. One thing it will see and report are the individual and total voltages of the replacement battery. It has no clue as to the actual capacity of the cell so it can't report the current charge status or the capacity or the pack.

The battery fail-safes (parameters loaded into the autopilot are set on the remaining charge or capacity of the battery. As such it is important that the BMS is trained to the replacement battery which is the purpose of the BMS Calibration you are about to perform.

In a nutshell this is what is accomplished by the BMS Calibration process-

1. **Draining** down the cells to as close to 3.0 volts and in the process completely draining the charge of the battery to 0%. At this point the BMS will automatically record this state of the battery referred to as the "Low End.
2. **Recharging** the battery to it's full capacity ("High End") in my case I used a 5200mAh battery.

Once this data is recorded the BMS does the math comparing the battery's current state with it's known capacity and streams that information through the com link to the controller which is the battery data you see on the HUD of the app. Depending on the version of FW you are

using the autopilot will trigger a fail-safe and Return to Launch (RTL) when these preset parameters have been reached.

If the BMS is not calibrated to the replacement battery it can not stream the charge status (remaining capacity) so you will not have the protection of the RTL fail-safe.

So now you know the HOW and WHY. Before we get started another “quick” word about equipment. As we proceed I will provide different methods in each of the calibration steps. Some will require additional equipment.

Solex Application: If you haven’t invested in Solex by now here is yet one more reason to help you justify shelling out $30. Solex provides a Battery Drain utility which is described in the process below.

Hobby Charger: To my way of thinking an absolute MUST have as it eliminates using your Solo to perform the battery drain.

Battery Tester: Again, if you are going to fly Solo the battery tester is as close to a requirement as you can get. The OEM Batteries were last produced in November 2015. As you are well aware they have or will soon reach their end of life as all LiPO batteries do. Flying OEM batteries now come with a risk of failing in flight as a weak cell can fail at any time. Testing your battery’s cell voltages and remaining capacity before each flight is paramount to safe flying. As an added bonus you will learn in the following section on BMS Calibration that testing the battery at different stages is very useful. The Madhacker Battery Tester is available at this site. Madhacker Store YOU NEED THIS TOOL!!!



STEP ONE: DISCHARGING THE MODIFIED BATTERY PACK

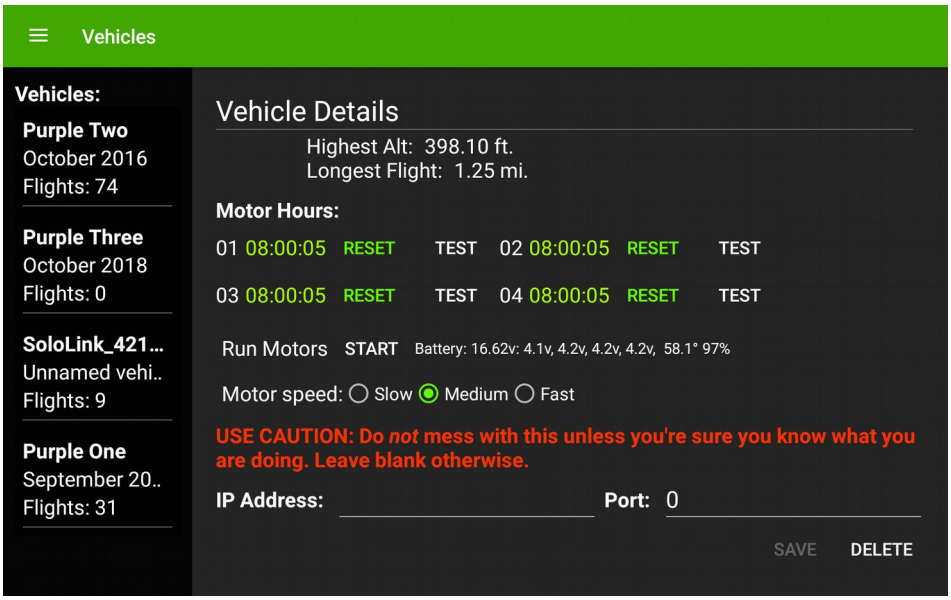
As was the case with the actual modification there are several choices in how you perform the individual steps in calibrating the BMS to the replacement battery. Be diligent NOT to discharge the battery beyond 12 volts (3.0 volts/cell) as irreparable damage can occur to to the battery.

Solex Method- (Data displayed assumes Green Cube on Open Solo FW 3.5 and above)* Attach the Battery Mod to Solo. Connect to the controller and bring up the Solex app. With props attached and the Solo in a safe location away from people and pets select **Vehicles** in the Menu.
Choose the Solo you are connected to and at the bottom of the page is a **Run Motors** utility.

You will see the Battery volts, individual cell volts, and temperature of the pack. Disregard the state of charge if it displays. That's old data left in the BMS memory. If your Solo has a gimbal be sure to mount the camera or you will burn out the motors in the gimbal.

Set Motor speed to **Fast.** Not to worry this is 25% throttle and not sufficient to launch the Solo. Click on **START** and the Solo will run the battery down to 12v or when any one of the cells hits 3.0v and cut off the motors. This can take hours depending on how much charge is in the battery.
WARNING!! After motors cut off you must power down the Solo to prevent battery drain beyond 12V.

NOTE: This is a viable discharge method but adds hours to your motors which needs to be taken into consideration.
*Some of the battery data shown in the screenshot will not be available on the stock autopilot running Solo FW.
GOOD NEWS! 3.7 FW release will add almost all of the capabilities to to the stock cube now available with the Green Cube.

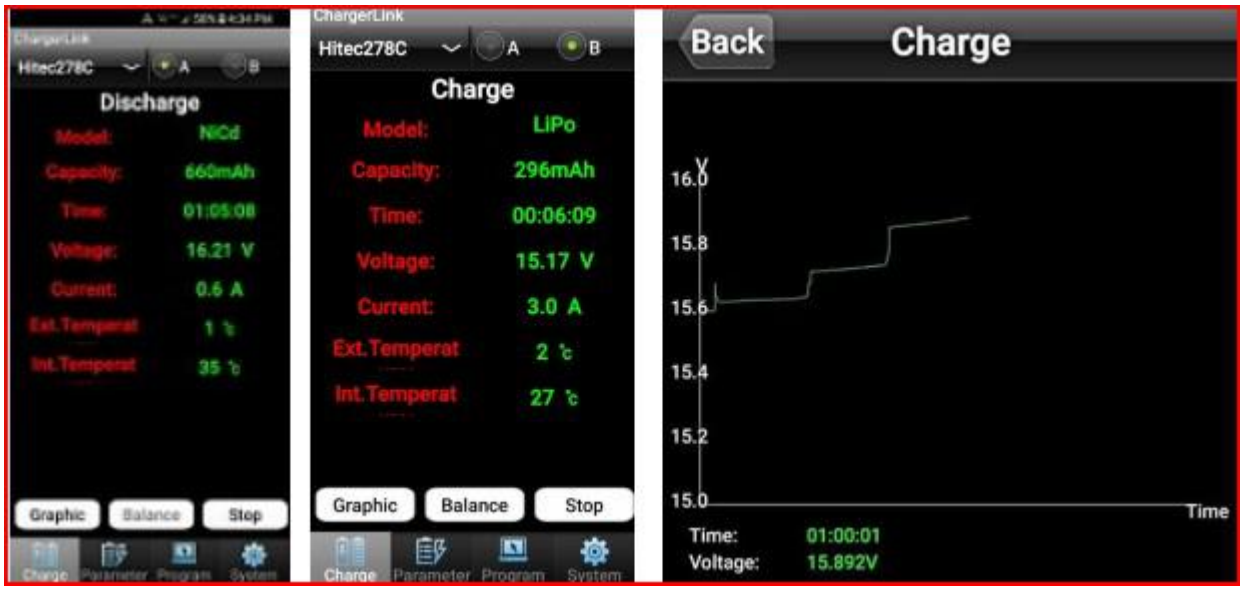


Hobby Charger Method-
Considering the wear and tear and safety risk using the Solo to discharge the battery I *highly* recommend that you consider the purchase of a hobby charger. They can be had for as little as $50 and make the discharge process both safe and well controlled. I purchased the HiTec X2. It's not the best one on the market but more than adequate for our needs. I can discharge and charge two batteries independently. I used a pair of crimp spades to adapt my charger's cable connection into the BMS.





Another nice feature of many chargers is the optional WiFi dongle (see photo) that connects to your mobile app so you can monitor the progress of the drain and charge cycle in progress.



Most LiPO battery charge/discharge modes will not set a discharge cut off below 14.0 volts. For this reason I use the NiCD Battery Mode on my charger and set the discharge rate at 1.0A and my discharge cut off at 13.5V. After the discharge cycle completes let the battery rest for at least one hour to let it balance out. At this point it is very advisable to connect your Solotool tester to the batter to ensure you have totally depleted the pack. Note that the individual cells are balanced within tolerances. The Des(ign) C(apacity): 5200mAh and Act(ual) C(apcity): 2781mAh are remnants of the last battery cycle. Now that we have “shown” the BMS the low end Rel(itive) Chg: 0% and Abs(olute) Chg: 0% it will compare these values to those achieved after the battery is fully charged.



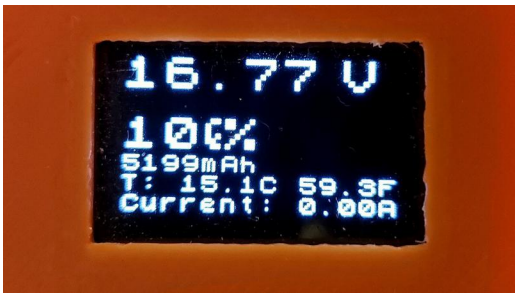


STEP TWO: CHARGING THE MODIFIED BATTERY PACK

Next is to connect the drained pack to either your Solo or Hobby charger and fully charge the drained pack. A quick word about charge settings. Use the LiPO battery mode set at 5200mAh/14.84v (4S) cutoff. Depending on the C rating of your replacement battery you can set the charge rate to 1-5 amps. The Tattu used in this tutorial has a 35C rating so I set my charger to a 4.0A rate. The battery manufacturer should provide a recommended charge rate. The Solo charger is set to 16.8v/3.3A

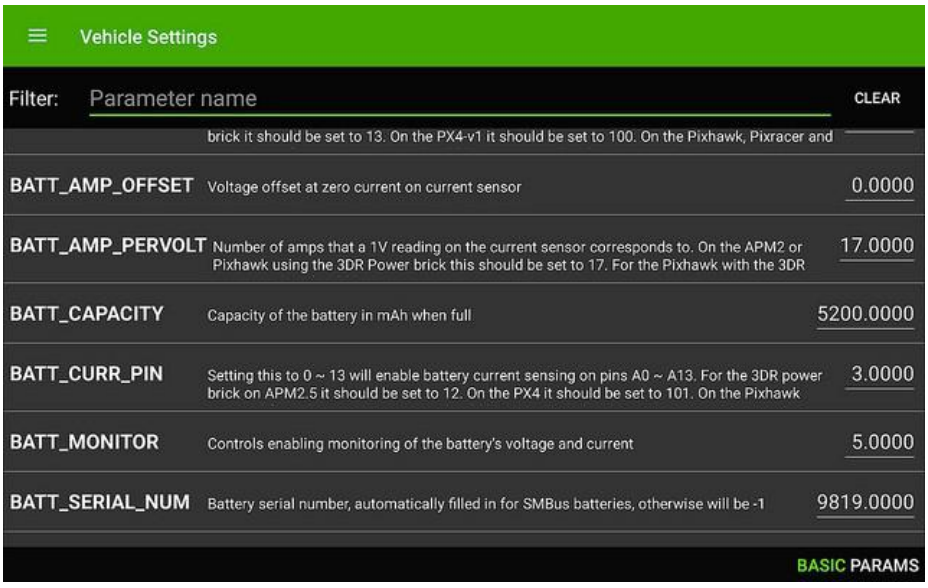


After the battery fully charges check the outcome. You should see something close to these measurements.



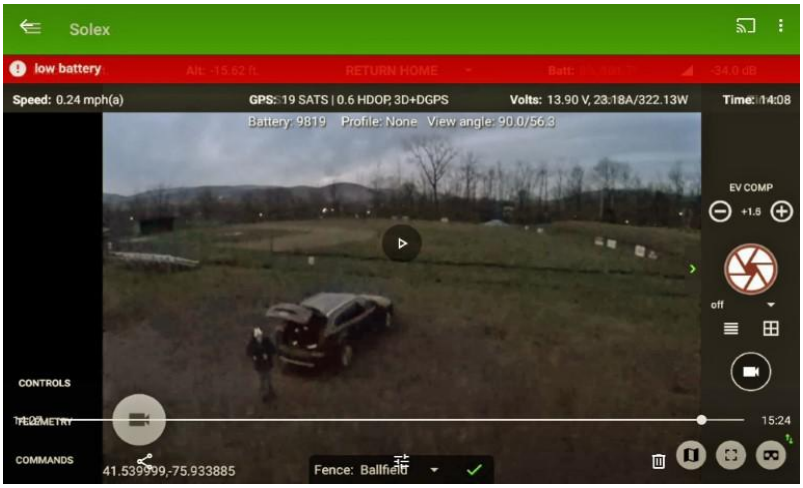


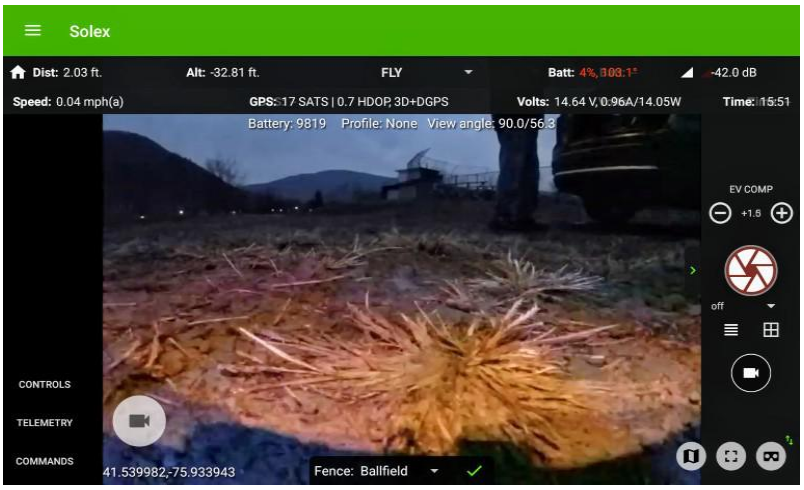
If you are still waiting for your tester to arrive then you can alternatively connect the battery to your Solo, boot it up and use the basic parameter utility in Solex **Vehicle Settings** searching for **BATT_CAPACITY** to see what your new mAh value is.



If you don't get to 5200mAh on the first round calibration fear not! Simply fly the Solo down to RTL then repeat the Drain/Charge. In most cases it takes about three BMS Calibration cycles to train the chip.

Here's a screen shot of my first flight on the Battery Mod. I flew it hard and fully loaded with Gimbal/GP (recording the whole time) being sure to Loiter for about 5 minutes of the total flight (Loiter uses more amps) to get a "normal" flight. As you can see I hit RTL at 14:08 minutes with 8% capacity and held it manually to 4%. I don't recommend flying the battery below 10-15% but wanted to see how long I could keep the new battery in the air. Landing at 4% got me 15:51 minutes of flight time.





So there you have it! I hope our tutorial has been helpful. If you have any concerns or questions please feel free to PM myself, Mike Lione, Jon Brotherton, or Chris Goodall anytime.

This has been a team effort by Carl Oliveria, Jon Brotherton, Chris Goodall, and myself. THANK YOU for all your hard work and expense to make our Solo Battery Mod for Dummies a reality.

One last word. If you found this Tutorial helpful be sure to spread it around to your fellow Solo Fliers. It can be found online and available for downloading at this link:

SOLO BATTERY MOD FOR DUMMIES

PRODUCTS PAGE-

So you read all the way to the end of this tutorial and have come to the conclusion that you rather not jump in. Here's some alternatives that have been made available since this document was provided back in 2018,

Purchase the Battery Mod completely assembled described in this tutorial:
Jon Brotherton assembled Battery Mod

More recently The Madhacker who developed the Solo Battery Tester used in this tutorial also offers the BMSOne which is completely configurable to both LiPo and LiIon battery chemistry. Additionally it can auto sense and adapt to both the 4S and 5S cell wiring.

BMSOne-

Madhacker's BMSOne manual and technical details:
Madhacker BMSOne/V2

Purchase the BMSOne (now sold only through JB Mod Parts), Plug and Play or board:
BMSOne Plug and Play

BATTERIES-

The battery pack incorporated into the Battery Mod is a high quality exact equivalent of the cell pack that was used in the original Solo battery and best suited for the battery mod:
Tattu-5200mAh 35C-4S1P LiPo Battery Factory Store
Tattu-5200mAh 35C-4S1P LiPo Battery Walmart

By using the BMSOne you can use higher capacity LiIon batteries for longer flight times:
Upgrade Energy-5600mAh 40A-4S2P LiIon Battery
Upgrade Energy-8000mAh 30A-5S2P LiIon Battery